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Intelligent Stock Portfolio Advisor

Introduction

Retail investors, financial advisors, and investment clubs often compare company health, price behavior, and market discussion across separate sources. This project combines those inputs in a local dashboard with transparent signals, a risk-profile allocation view, simulated order entry, and an attribution report. The aim is to make assumptions behind a portfolio idea visible and easy to discuss.

The prototype uses eight example tickers and generated daily prices from 2023 through 2025. It demonstrates a reproducible research workflow but does not issue advice on real securities, store client portfolios, or connect to a brokerage.

Problem Statement

Company signals combine fundamental percentiles for revenue growth, return on equity, valuation, and debt with trend and momentum indicators, including 20- and 50-session averages, RSI-14, and MACD. The default composite uses 40% fundamentals, 35% technicals, and 25% text sentiment. Factors are scaled to a common 0-to-1 range; BUY begins at 0.60, SELL is 0.40 or lower, and intermediate scores map to HOLD.

An optional per-ticker LSTM forecasts the next closing price from 60-session windows using a chronological 80/20 split and a previous-close baseline. Optional FinBERT classifies financial text as positive, negative, or neutral; a small lexicon is the default fallback. Training code and model inference are included, but no trained LSTM weights are bundled.

Results and Discussion

The demo contains 6,264 synthetic OHLCV rows for eight stocks across 783 weekdays from 2023 to 2025, plus eight fundamental profiles and 96 fabricated discussion posts. The application includes six views for overview, signals, optimization, paper trading, attribution, and forecasts. Portfolio allocation uses annualized sample means and covariance, a long-only constraint, and a 40% per-stock cap. Paper orders use the last sample close. Since the inputs are synthetic and no LSTM has been trained, the report makes no real-market return or forecast-accuracy claim.

Platform Summary

Overview charts prices and indicators; Stock signals explains factor scores; the optimizer proposes allocations; Paper trading tracks simulated cash and holdings; Monthly report attributes a fixed model portfolio; and LSTM forecast displays saved per-ticker outputs and holdout errors.

Signal factor	Default weight	Basis
Fundamentals + technicals	75%	Quality, valuation, trend, and momentum
News and social sentiment	25%	FinBERT option; lexicon default

Default composite: 40% fundamentals + 35% technicals + 25% text sentiment. For a ticker with a saved LSTM forecast, the forecast contributes 10% and the other factors are rescaled. Risk profiles change the variance penalty, and a 40% position cap limits single-stock concentration.

Signal and Allocation Rules

```
signals, source = build_recommendations(prices, fundamentals, discussions)
allocation, stats = optimize_portfolio(training_prices, risk_profile)
report = model_monthly_attribution(prices, allocation)
```

```
BUY >= 0.60  |  HOLD > 0.40 and < 0.60  |  SELL <= 0.40
MPT: long-only; sum of weights = 1; each stock <= 40%
```

Risk profiles adjust the variance penalty in the mean-variance optimizer. The monthly report estimates its allocation using data available before the report period, then holds it fixed. Paper orders execute without fees or slippage, and the in-memory account resets when the browser session ends. The prototype does not model dividends, taxes, corporate actions, market impact, client suitability, or persistent account storage. Synthetic inputs cannot establish whether the models generalize to real market data.

Conclusion

This project combines multi-factor stock screening, optional sequence and text models, risk-based allocation, paper trading, and return attribution in a local prototype. Factor definitions and allocation constraints are inspectable, while paper trading avoids real orders. Synthetic inputs and untrained optional models make it a learning demonstration rather than an investment service. A production system would need licensed point-in-time data, walk-forward validation, realistic execution costs, secure account persistence, and advisor review controls.

Reference

Markowitz, H. (1952). Portfolio Selection. The Journal of Finance, 7(1), 77-91. https://doi.org/10.1111/j.1540-6261.1952.tb01525.x. Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation, 9(8), 1735-1780. https://doi.org/10.1162/neco.1997.9.8.1735. Araci, D. (2019). FinBERT: Financial Sentiment Analysis with Pre-trained Language Models. arXiv:1908.10063. https://arxiv.org/abs/1908.10063.